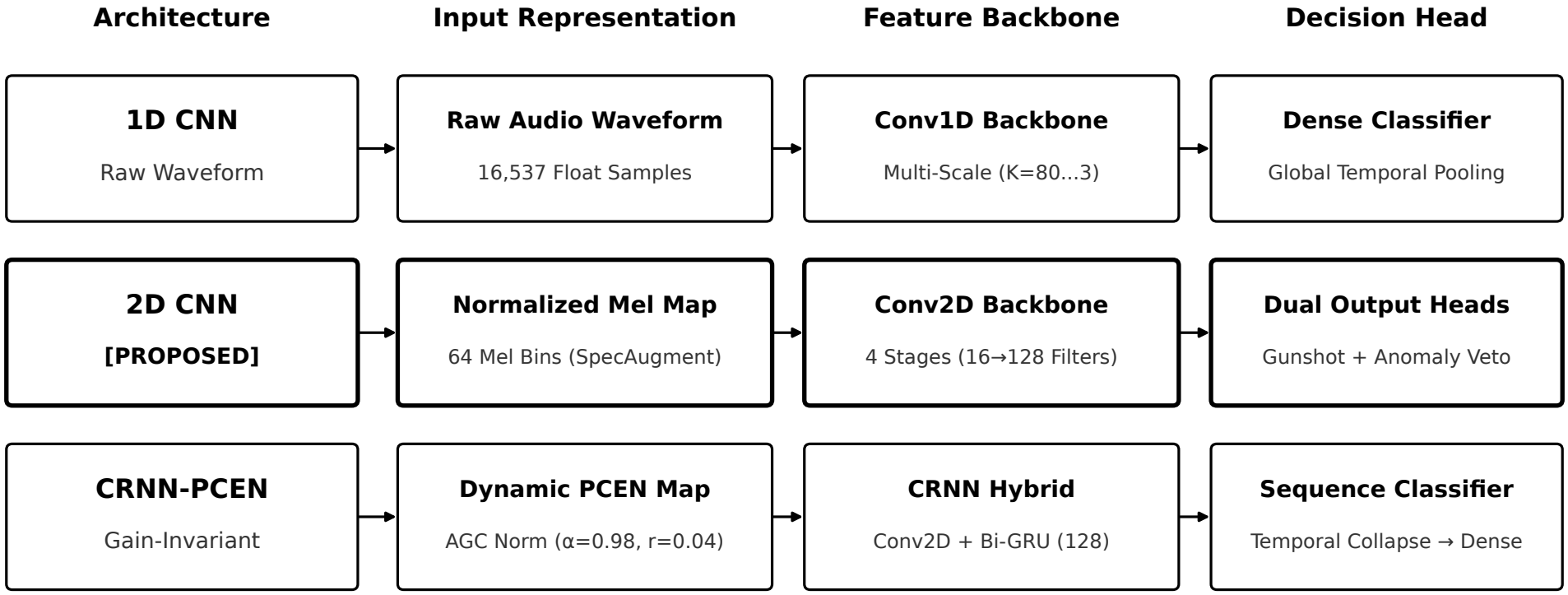
Architecture
Input Representation
Feature Backbone
Decision Head
1D CNN
Raw Waveform
Raw Audio Waveform
16,537 Float Samples
Conv1D Backbone
Multi-Scale (K=80...3)
Dense Classifier
Global Temporal Pooling
2D CNN
[PROPOSED]
Normalized Mel Map
64 Mel Bins (SpecAugment)
Conv2D Backbone
4 Stages (16→128 Filters)
Dual Output Heads
Gunshot + Anomaly Veto
CRNN-PCEN
Gain-Invariant
Dynamic PCEN Map
AGC Norm (α=0.98, r=0.04)
CRNN Hybrid
Conv2D + Bi-GRU (128)
Sequence Classifier
Temporal Collapse → Dense